

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26057

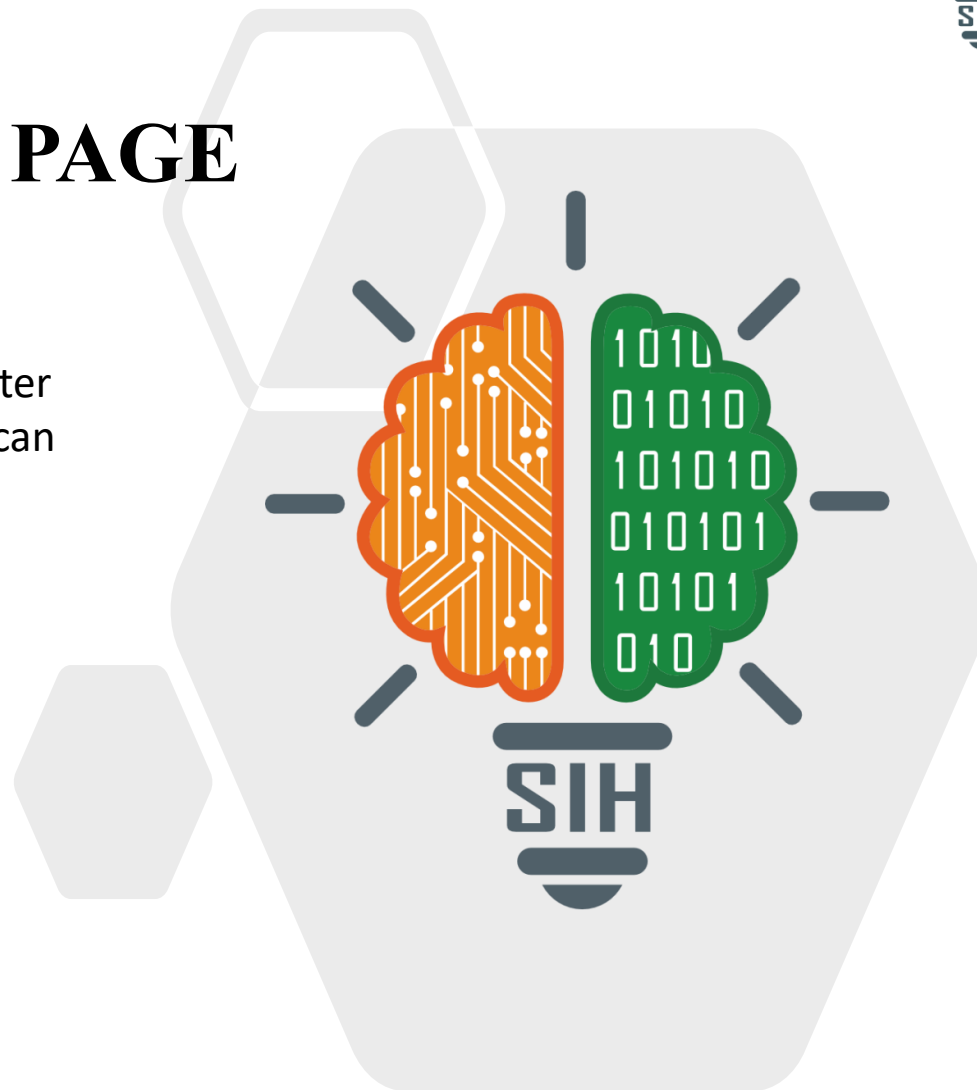
Problem Statement Title- AI-Powered Automated Underwater
Marine Debris and Anomaly Detection System using Side-Scan
Sonar Imagery

Theme- Disaster Management

PS Category- Software

Team ID- <Team ID>

Team Name (Registered on portal)- Vashishta



IDEA TITLE: SonarSentinel



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

Proposed solution

SonarSentinel is an end-to-end AI pipeline + map dashboard that reads raw side-scan sonar logs (.xtf, GeoTIFF, image + navigation CSV), detects man-made debris and ghost nets, and reports each hazard's GPS position, size, class and a 0–100% confidence.

How it addresses the problem

- Replaces slow manual review of sonar logs with automatic detection shown live on a map
- Separates debris from rocks, sand ripples and acoustic shadows using sonar physics + AI
- Handles speckle noise, varying pixel resolution, data dropouts and heave / pitch / roll
- Runs offline on a laptop or on an AUV / marine-drone edge computer

Innovation and uniqueness

- GPS computed from sonar ping headers: every detection geotagged with an uncertainty estimate
- Finds ghost nets without a ghost-net dataset: synthetic nets + anomaly detection + expert review loop
- Physics-aware, calibrated and explainable confidence (shadow-geometry check + score breakdown)
- Edge-ready (INT8 TensorRT / OpenVINO) with open outputs: JSON, CSV, GeoJSON, KML

Example output (illustrative)

Class: ghost net

Confidence: 87.4% · HAZARD

Position: 13.084120 N, 80.312750 E

Size: 6.2 m × 3.1 m · Depth: 18.5 m

Position uncertainty: ± 4.2 m

Source: line_07.xtf, starboard side, pings 10,398–10,466

Delivered as JSON / CSV / KML for dive and ROV teams

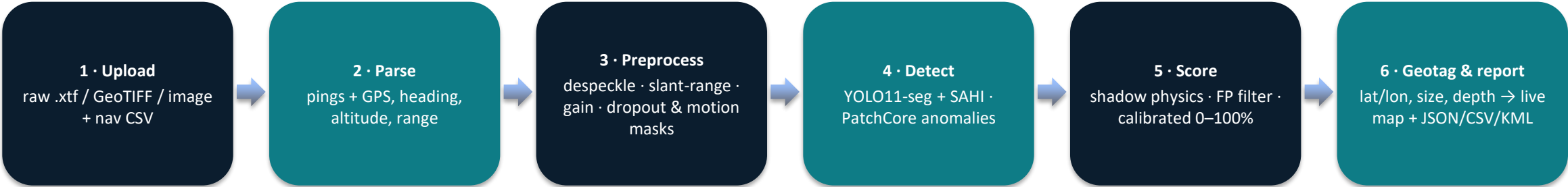
Workflow

Upload sonar log → Detect → Score → Geotag → Live map + report

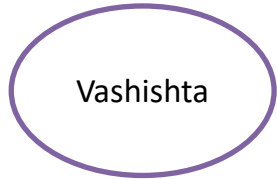
- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

Layer	Technologies to be used
Sonar I/O & geo	Python 3.11 · pyxtf (XTF ping headers) · GDAL / rasterio · pyproj
Signal processing	OpenCV · NumPy / SciPy: Lee despeckle, slant-range correction, gain normalisation, dropout & motion masks
AI models	YOLO11-seg + SAHI (detection & masks) · PatchCore / anomalib (unknown objects) · LightGBM false-positive filter · isotonic calibration
Application	FastAPI (REST + WebSocket live streaming) · React + Leaflet (offline maps) · SQLite → PostGIS
Edge & deployment	ONNX · TensorRT (NVIDIA Jetson) · OpenVINO INT8 · Docker

Methodology and process for implementation



- Long sonar logs are processed in overlapping chunks and streamed to the map in real time (WebSocket)
- Expert confirm / reject decisions become new training data; models are exported to ONNX for edge devices



FEASIBILITY AND VIABILITY

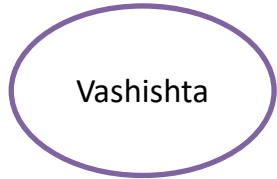


- *Analysis of the feasibility of the idea*
- *Potential challenges and risks*
- *Strategies for overcoming these challenges*

Analysis of feasibility

- Fully open-source stack (AGPL-3.0): runs on a GPU laptop, CPU-only, or a Jetson-class edge computer
- Public data available now: AI4Shipwrecks (286 labelled SSS images), mine-detection SSS dataset (1,170 images, CC BY 4.0), NOAA hydrographic surveys with navigation
- Six-sprint build plan with weekly milestones; requirements, architecture and test plan already defined

Potential challenges and risks	Strategies for overcoming them
No public labelled ghost-net sonar dataset	Synthetic ghost-net generator on real seabed + anomaly detection + expert review queue that collects real labels
Rocks, sand ripples and shadows cause false alarms	Shadow-geometry check, shape / texture false-positive filter, hard-negative training, calibrated alert tiers
Different sonars, frequencies and resolutions	Resample to a fixed 10 cm / pixel, gain normalisation, sonar-specific augmentation, fine-tuning on NIOT samples
Position errors (towfish layback, heading, GPS gaps)	Layback correction, navigation smoothing, per-detection uncertainty, validation against charted wrecks
Limited compute on marine drones	Compact YOLO11 model with INT8 TensorRT / OpenVINO; heavy steps (anomaly model, mosaic) run on shore



IMPACT AND BENEFITS



- *Potential impact on the target audience*
- *Benefits of the solution (social, economic, environmental, etc.)*

Potential impact on the target audience

- NIOT / MoES survey teams: automated screening of survey logs; analysts review only flagged hazards
- Port, coastal and disaster authorities (Navy, Coast Guard): rapid post-cyclone / post-tsunami hazard maps
- Fisheries departments, NGOs and dive / ROV teams: exact GPS positions for ghost-net recovery

Benefit	Details
Environmental	Faster ghost-net discovery → less entanglement of fish, turtles and dolphins and less reef damage; supports UN SDG 14 (Life Below Water)
Disaster management	Rapid mapping of submerged debris in ports and shipping channels after cyclones, floods and tsunamis
Economic	Fewer analyst hours per survey (target ≥ 70% time saving, to be measured); less propeller and fishing-gear damage; faster channel reopening
Social & operational	Same-day hazard reports with GPS for field teams; on-board alerts; offline and on-premise, so sensitive seabed data stays in-country
Strategic	Indigenous, open, edge-deployable capability aligned with MoES / NIOT ocean-technology programmes

RESEARCH AND REFERENCES



• *Details / Links of the reference and research work*

- [1] Z. Miao et al., "GhostNetZero: AI for detecting marine ghost nets," Microsoft AI for Good Lab, 2025.
- [2] A. V. Sethuraman et al., "Machine learning for shipwreck segmentation from side scan sonar imagery: Dataset and benchmark," Int. J. Robotics Research, 44(3), 2025. doi:10.1177/02783649241266853
- [3] N. Pessanha Santos et al., "Side-scan sonar imaging data of underwater vehicles for mine detection," Data in Brief, 53, 2024. doi:10.1016/j.dib.2024.110132
- [4] G. Huo, Z. Wu, J. Li, "Underwater object classification in sidescan sonar images using deep transfer learning and semisynthetic training data," IEEE Access, 8, 2020.
- [5] K. Roth et al., "Towards total recall in industrial anomaly detection" (PatchCore), CVPR 2022.
- [6] F. C. Akyon, S. O. Altinuc, A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," ICIP 2022.
- [7] C. Guo et al., "On calibration of modern neural networks," ICML 2017.
- [8] K. Richardson, B. D. Hardesty, C. Wilcox, "Estimates of fishing gear loss rates at a global scale," Fish and Fisheries, 20(6), 2019. doi:10.1111/faf.12407
- [9] P. Blondel, The Handbook of Sidescan Sonar, Springer/Praxis, 2009.
- [10] Data & tools: NOAA NCEI hydrographic survey data; Ultralytics YOLO11; pyxtf (github.com/oysstu/pyxtf); anomalib.